Clinical Immunology Communications 2 (2022) 91–97

Contents lists available at ScienceDirect

Clinical Immunology Communications

journal homepage: www.elsevier.com/locate/clicom

Full Length Article

Covid-19 Vaccine safety and adverse event analysis from Pakistan

Syed Shahzad Hasan[a,b], Abdur Rashid[c], Sundus Osama[d], Zia Ul Mustafa[e], Hamid A. Merchant[a,f,*]

[a] Department of Pharmacy, School of Applied Sciences, University of Huddersfield, Huddersfield, UK
[b] School of Biomedical Sciences and Pharmacy, University of Newcastle, Callaghan, Australia
[c] Pakistan National Pharmacovigilance Centre, Division of Pharmacy Services, Drug Regulatory Authority of Pakistan, G9/4 Islamabad, Pakistan
[d] Tehsil Headquarter Hospital, Kuldana Road, Murree, Pakistan
[e] Department of Pharmacy Services, District Headquarter Hospital, Pakpattan, Sahiwal, Pakistan
[f] Health Services Academy, Chak Shahzad, Islamabad 44000, Pakistan

ARTICLE INFO

Keywords:
Pharmacovigilance
VAERS
Yellow-Cards
COVID
SARS-COV-2
Coronavirus
Covid vaccine

ABSTRACT

Covid immunization commenced on 2nd Feb 2021 in Pakistan and as of 7th Sep 2021, over 84 million vaccine doses were administered in Pakistan, of which 72% procured by the government, 22% received through Covax and 6% were donated. The vaccines rolled out nationally included: Sinopharm, Sinovac and CanSinoBIO (China), AstraZeneca (UK), Moderna and Pfizer (USA), Sputnik (Russia), and PakVac (China/Pakistan). About half of the eligible population in Pakistan (63 m) had received at least one dose of Covid vaccine as of Sep 2021. Pakistan National Pharmacovigilance Centre (PNPC) in coordination with WHO, MHRA and Uppsala Monitoring Centre (UMC) established pharmacovigilance centers across Pakistan. The Covid vaccine AEFIs in Pakistan were mainly reported via NIMS (National Immunization Management System), COVIM (Covid-19 Vaccine Inventory Management System), 1166 freephone helpline and MedSafety. There have been 39,291 ADRs reported as of 30th Sept 2021, where most reported after the first dose ($n = 27{,}108$) and within 24–72 h of immunization ($n = 27{,}591$). Fever or shivering accounted for most AEFI (35%) followed by injection-site pain or redness (28%), headache (26%), nausea/vomiting (4%), and diarrhoea (3%). 24 serious AEFIs were also reported and investigated in detail by the National AEFI review committee. The rate of AEFIs reports ranged from 0.27 to 0.79 per 1000 for various Covid vaccines in Pakistan that was significantly lower than the rates in UK (~4 per 1000), primarily atrributed to underreporting of cases in Pakistan. Finally, Covid vaccines were well tolerated and no significant cause for concern was flagged up in Pakistan's Covid vaccine surveillance system concluding overall benefits outweighed risks.

Key Points

- There were eight Covid vaccine products being rolled out in Pakistan including conventional vaccine formulations (whole inactivated virus vaccine), mRNA and Adv vector-based vaccines to meet national vaccine demands.
- There have been 39,291 AEFIs reported, comprising most of minor adverse reactions usually expected following immunization, including 24 serious AEFIs.
- The overall AEFI reporting rate ranged from 0.27 to 0.79 per 1000 for various Covid vaccine products in Pakistan that was significantly lower than the rates reported in UK (~4 per 1000), primarily due to underreporting of cases in Pakistan.
- Covid vaccines were well tolerated overall and no significant cause for concern was flagged up in Pakistan's Covid vaccine surveillance system concluding overall benefits outweighed risks.

Introduction

The national pharmacovigilance activities (PA) were established in Pakistan over 25 years ago. However, its effectiveness was not at par with the international regulatory standards. This was mainly due to the lack of trained human resources, infrastructure, poor reporting system, and feeble national pharmacovigilance guidelines [1]. The Drug Regulatory Authority of Pakistan (DRAP) was established in 2012 as a major reform in the drug regulatory infrastructure of Pakistan following the incident at the Punjab Institute of Cardiology, Lahore, where the drug contamination caused more than 200 deaths and left at least 1000 seriously ill [2].

DRAP is an autonomous body established under the administrative control of 'The Ministry of National Health Services, Regulation and Coordination' Government of Pakistan to enforce The Drug Act 1976

* Corresponding author.
E-mail address: hamid.merchant@hud.ac.uk (H.A. Merchant).

https://doi.org/10.1016/j.clicom.2022.05.003
Received 3 March 2022; Received in revised form 12 May 2022; Accepted 13 May 2022

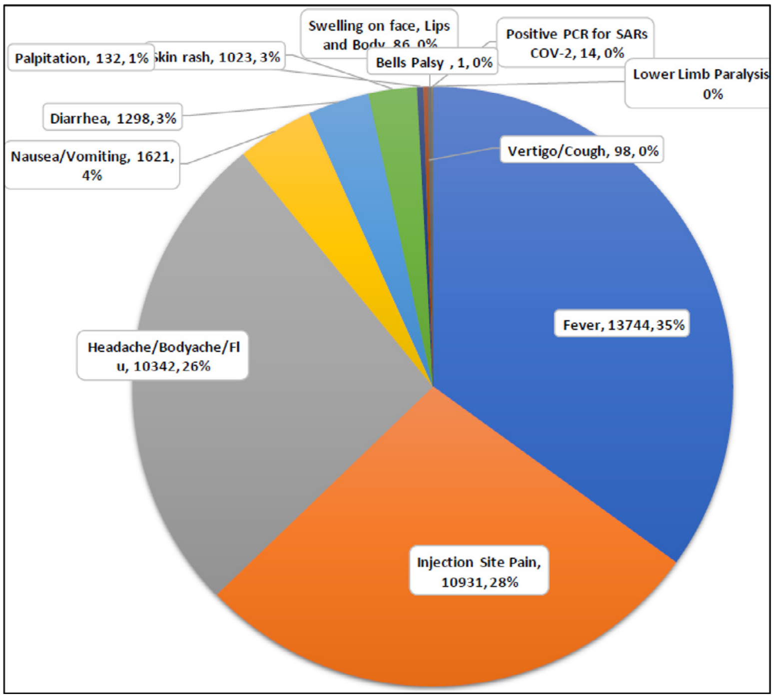

Fig. 1. Adverse events following Covid immunisation reported in Pakistan.

Table 1
Adverse events following Covid immunisation reported in Pakistan from various Covid vaccine products.

S. No	Vaccine Name	Active Ingredient	Doses Administered	AEFI Cases	AEFI per 1000
1.	Sinopharm	Covid-19 vaccine inactivated (Vero) HB02	25,895,889	16,291	0.63
2.	CoronaVac-Sinovac	Covid-19 vaccine inactivated (Vero) CZ02	39,349,854	15,243	0.39
3.	CanSinoBIO	Covid-19 vaccine NRVV Ad5 (AD5-nCoV)	2841,263	2234	0.79
4.	Vaxzevria-AstraZeneca	Covid-19 vaccine NRVV Ad (ChAdOx1 nCoV-19)	2620,094	1638	0.63
5.	Spikevax - Moderna Covid-19 Vaccine	Single-stranded, 5′-capped messenger RNA (mRNA) encoding the SARS-CoV-2 spike (S) protein	3452,175	1897	0.55
6.	PakVac	Covid-19 vaccine NRVV Ad5 (AD5-nCoV)	3173,100	1677	0.53
7.	Pfizer	Single-stranded, 5′-capped messenger RNA (mRNA) encoding the SARS-CoV-2 spike (S) protein	1114,319	298	0.27
8.	Sputnik V	Recombinant adenovirus particles (Adv 26 first dose, Adv 5 second dose) containing SARS-CoV-2 protein S gene	198,890	13	0.07

and facilitate the trade of therapeutic goods throughout the country since its establishment. DRAP is administered by thirteen directors, including a Director of Pharmacy Services responsible for developing and promoting pharmacy services in the country. One of the key responsibilities of the pharmacy services division is to monitor pharmacovigilance activities throughout the country. The division also established the Pakistan National Pharmacovigilance Centre (PNPC) in coordination with the World Health Organization (WHO) and Uppsala Monitoring Centre (UMC). This also led to the establishment of pharmacovigilance centers at provincial levels for the subsequent ADRs (adverse drug reactions) reporting, disseminating post-marketing surveillance data and global safety updates [1]. Owing to all these ongoing efforts of PNPC-DRAP, Pakistan became the 134th full member of the WHO program in November 2018. In addition, Pakistan's new National Pharmacovigilance Guidelines were published in 2019 to effectively implement pharmacovigilance in the country [3]. The newer systems meant that the ADRs can now be easily reported online to the PNPC through MED Vigilance e-Reporting System.

'Med Safety' app for ADR reporting

Recently during Covid-19 pandemic, DRAP has also launched a smartphone application called 'Med Safety' that was made available to download via Google's Play Store and Apple's App Store for Android and iOS-based smartphones and other hand-held devices. The app allows the online reporting of any suspected side effects from allopathic, biological, herbal, and homoeopathic products. This application aimed to offer a convenient means of ADR reporting and encouraged the active participation of all healthcare professionals and patients to help improving medication safety.

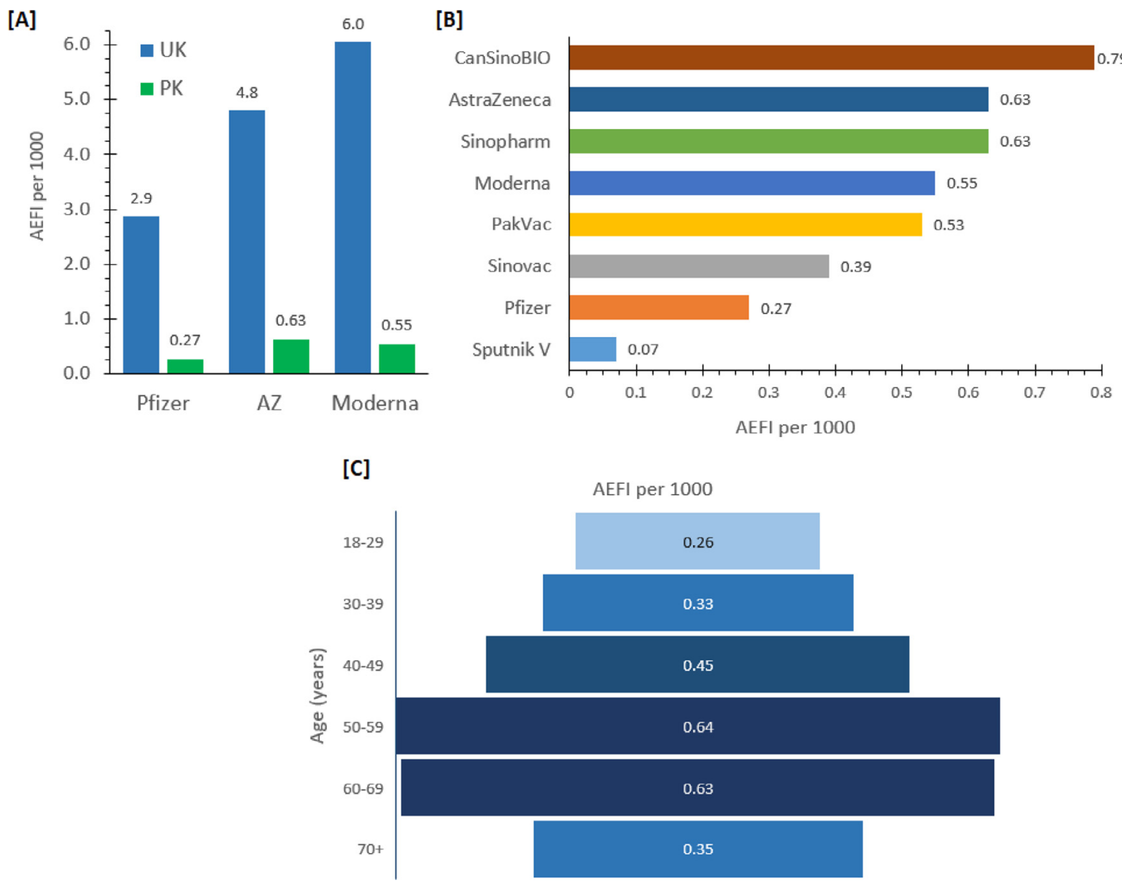

Fig. 2. **[A]** Adverse events/1000 following Covid immunisation for Pfizer, AstraZeneca and Moderna vaccines reported in Pakistan vs UK, **[B]** Adverse events/1000 following Covid immunisation for various vaccine products reported in Pakistan, **[C]** Adverse events/1000 following Covid immunisation in various age groups in Pakistan.

The application was originally developed by the MHRA (the Medicines and Healthcare products Regulatory Agency in the UK), enabling a direct reporting of all ADRs to the WHO. The new smartphone app in Pakistan was a joint effort between DRAP, WHO, UMC, and the MHRA to allow easy and instant ADR reporting by health professionals and the public in Pakistan [4]. This user-friendly application also offered an offline mode that allowed users to submit reports on adverse drug reactions even when they did not had an active internet connection. As soon as the device had access to the internet, the stored ADRs were automatically submitted. An immediate acknowledgement of the receipt of the ADR report was automatically sent to the reporter. Moreover, users can also access previously submitted reports, review medicine safety updates, and create their personalized watch-list of medications. They can also subscribe to receive regular updates and alerts on medical safety and adverse events [5].

Covid vaccine pharmacovigilance procedure in pakistan

The Covid vaccine adverse events were mainly reported via NIMS (National Immunization Management System), COVIM, and the 1166 freephone helpline. The reports from other channels, including social media, were also logged, and adverse events were noted from weekly VPD Surveillance reports, and EPI MIS reports from districts and provinces. There have been over four thousand Covid vaccine centers established across Pakistan, where each centre was equipped with dedicated areas and bed spaces for people to rest for 30 min following immunization. If any AEFI was observed during this period, information was immediately collected in an AEFI reporting form at the vaccination centre.

a) National Immunization Management System (NIMS)

NIMS has a built-in section to report any adverse event observed following immunisation (AEFI). After receiving a Covid jab, a short text message (SMS) is automatically generated within 48–72 h to the registered mobile number of the vaccinated subjects through NIMS system offering an opportunity to report any adverse reactions to the vaccine. The first SMS is sent to enquire if the subject has observed any adverse event that can be responded back by a direct 'Yes or No' response using a simple text message. If answered 'Yes', a second SMS will follow listing possible side effects with an IVR code, and a response can be given by a direct message by selecting an appropriate IVR code. The system also offers adverse event reporting via the internet through NIMS web portal for those who wish or prefer to do it this way.

a) Covid-19 Vaccine Inventory Management System (COVIM).

The designated hospital staff also reported all serious adverse events following immunisation that led to hospitalisation through COVIM.

a) 1166 telephonic helpline

1166 helpline was staffed with a team of operators and healthcare professionals specially trained in adverse events reporting and Covid-19 emergency regulations. They recorded all suspected Covid vaccine adverse events directly into the vaccine adverse event reporting system for further processing and investigations.

a) Other Channels

AEFIs reported through social media and direct reports from healthcare facilities to the district, provincial, and national focal persons for pharmacovigilance, were also registered.

a) EPI MIS and VPD Surveillance Report

All AEFI data from the district to provinces were shared weekly with the federal EPI through EPI MIS and Excel spreadsheets.

a) Med-Safety mobile app

Adverse events were also reported via Med-Safety mobile applications available across Android and iOS-based devices. The healthcare professionals and citizens were able to directly report adverse events through the mobile app.

a) Vigiflow-DRAP

All the AEFI reported through NIMs, COVIM, 1166, and weekly EPI Reports were entered into Vigiflow. To date, all data entry process was done at the federal EPI level.

Covid vaccines in Pakistan

There had been a total of 84.6 million vaccine doses arrived in Pakistan as of 7th September 2021; of which 61.2 million doses (72%) were procured by the Pakistan government, 4.7 million doses (6%) received as a donation, and 18.7% million (22%) received through Covax initiative. The vaccines received through Covax initiatives included Sinopharm (6.1 million doses), AstraZeneca (3.4 million doses), Moderna (5.5 million doses), and Pfizer (3.7 million doses). The other Covid vaccines made available in Pakistan included Sinovac, CanSinoBIO, PakVac, Sputnik, and Janssen (Johnson & Johnson).

Pakistan deployed a number of Covid vaccine products to meet the national immunisation demand amidst Covid-19 pandemic. This included Covid vaccines formulated using different technologies with anticipated differences in efficacy and safety between vaccine formulations in different populations. Therefore, vaccine surveillance and monitoring data was of utmost importance in settings like Pakistan, where a range of vaccine formulations were used nationally. The Covid vaccines rolled out in Pakistan included the following types:

a) Inactivated whole virus vaccines: These vaccines included ***Sinopharm*** and ***Sinovac*** Covid vaccines made using conventional whole inactivated virus technology and were adjuvanted with alum. The Sinopharm was based on 19nCoV-CDC-Tan-HB02 (HB02) strain, whereas Sinovac used the SARS-CoV-2 CN02 strain. The vaccine formulation and manufacturing techniques were similar to previously used conventional vaccines, such as the seasonal flu vaccine. These vaccines required two doses to complete the immunization course.
b) Viral vector vaccines: These vaccines used silenced adenovirus as a vector to deliver Covid spike genes to produce an immune response following immunization. These vaccines included Covid vaccines AstraZeneca (UK), Janssen/J&J (USA), Sputnik (Russia), and CanSinoBIO (China). ***AstraZeneca***, and ***Sputnik V*** required two doses, whereas ***Janssen (J&J)*** and ***CanSinoBIO*** were single-dose Covid vaccines. CanSinoBIO Covid vaccine was also clinically trialled locally in the Pakistani population for evaluating its safety and efficacy against locally prevalent strains. ***PakVac*** was the local version of the CanSionBIO vaccine that was imported in bulk and locally processed by the Pakistan's National Institute of Health's vaccines and biological products manufacturing facility in Islamabad.
c) mRNA vaccines: These vaccines were formulated using mRNA of the spike protein encapsulated in a proprietary solid-lipid nanoparticulate delivery system. These products included **Moderna** (USA) and **Pfizer** (USA) Covid vaccines.

Covid = vaccines administered in Pakistan

Of 125,853,762 eligible populations, about 63 million (50%) in Pakistan have had one or two doses of Covid vaccine as of 30th September 2021. About 48 million had received their first dose, and about 15 million had received two doses of Covid vaccines. Of these, 36 million subjects in Punjab, 13 million in Sindh, 9 million in KPK, 1.6 million in Capital city, 1.9 million in AJK, 1.1 million in Balochistan, and 0.7 million in GB received partial or full vaccination. More males were fully vaccinated (64%) than the females (36%) in Pakistan.

Table 2
AEFI per 1000 by demographic, geographical location, and doses.

	Doses	AEFI cases	AEFI per 1000
Gender			
Male	34,292,625	27,379	0.80
Female	23,532,392	12,542	0.53
Age			
18–29	19,540,656	5122	0.26
30–39	19,211,311	6414	0.33
40–49	15,737,848	7156	0.45
50–59	11,976,932	7721	0.64
60–69	6833,922	4326	0.63
70+	3390,506	1195	0.35
Healthcare providers	2533,032	7358	2.90
Provinces			
Punjab	44,534,423	18,471	0.41
Sindh	15,481,860	9165	0.59
KPK	12,160,692	5519	0.45
Capital	1832,207	861	0.47
AJK	2187,489	2742	1.25
GB	740,365	589	0.80
Balochistan	1515,813	1944	1.28
Vaccine Dose			
Dose 1	47,897,380	27,108	0.57
Dose 2	15,193,943	12,813	0.84

Covid vaccines ADRs and safety analysis in Pakistan

Since the beginning of the Covid immunization drives in Pakistan in February 2021, 39,291 ADRs have been reported from Covid vaccines up until 30th September 2021 in Pakistan. Most of the vaccinated people developed AEFI after the first dose (n = 27,108) and within 24 to 72 h (n = 27,591) following the vaccine administration. amongst AEFI reported for various Covid vaccines, fever, or shivering accounted for most AEFI as 35% (n = 13,744) out of the total were of this nature (Fig. 1). Injection site pain or redness was another common AEFI to Covid vaccine that contributed 28% (n = 10,931) of total ADRs. Headache were reported in 26% (n = 10,342) followed by nausea or vomiting 4% (n = 1621) and diarrhoea 3% (n = 1298). From eight registered Covid vaccines under DRAP, the highest AEFI per 1000 was reported for CanSinoBIO (0.79 AEFI per 1000), followed by Sinopharm (0.63), and Oxford-AstraZeneca (0.63). The lowest AEFI was reported for Sputnik (0.07 AEFI per 1000) and Pfizer (0.27 AEFI per 1000) (Table 1, Fig. 2B).

Table 2 presents the AEFI per 1000 vaccinated subjects according to demographics, geographical location, and doses received. AEFI per 1000 was higher in males (0.80 AEFI per 1000) and 50–69 years old (0.64 AEFI per 1000) (Fig. 2C). Healthcare professionals reported a considerably high number of AEFI that is 2.90 AEFI per 1000 vaccinated people. Regarding geographical location, the highest AEFI was reported by people vaccinated in Balochistan (1.28 AEFI per 1000). Interestingly, the AEFI rate was higher following the second dose (0.84 per 1000) than the first dose (0.57 per 1000).

Twenty-four ADRs were classed as serious. All serious AEFI were investigated in detail by the National AEFI review committee that concluded nineteen cases as coincidental (unrelated to the vaccine), whereas five serious cases were still under investigation at the time of writing this article; refer to Table 3 for details of all serious AEFIs reported to date in Pakistan. Most of the adverse events were reported through NIMS (about 91%), whereas about 8% of reports were from the 1166 helpline, and the rest came from other sources for instance as social media. Most of the adverse events (~85%) were reported within 72 h post-immunisation, whereas about 15% of events were reported

Table 3
List of serious adverse events reported following Covid vaccines in Pakistan as of 7th September 2021.

EPID#	Jurisdiction	Adverse Event	Age	Diagnosis	Vaccine type & dose	Reporting Channel	Status	Conclusion by the national AEFI review committee
001	Mirpur, AJK	Dizziness and Unconsciousness	58	Acute Coronary Syndrome	1st Dose Sinopharm	COVIM	Alive	Coincidental
002	Islamabad, Capital Territory	Severe vomiting and Chest discomfort & pain Heart attack	74	Double vessel coronary artery disease (DVCAD)	1st Dose Sinopharm	Director-General	Alive	Coincidental
003	Multan, Punjab	Unconscious and Lower Limb Paralysis	35	GBS	2nd Dose Sinopharm	Reported by Visiting Consultant through WhatsApp to AEFI Consultant	Alive	Coincidental
004	Peshawar, KPK	Heaviness of Tongue, Weakness left side of body, Fever, Severe headache, High Blood pressure	72	CVA	2nd Dose Sinopharm	WR WHO	Alive	Coincidental
005	Peshawar, KPK	Severe Headache, Heaviness of Tongue, High-Grade fever, Diplopia	60	TIA	1st Dose Sinopharm	WR WHO	Alive	Coincidental
006	Rawalpindi, Punjab	Slurring of speech, Severe Headache, High Blood pressure, Right Side body weakness	65	TIA	2nd Dose Sinopharm	WR WHO	Alive	Coincidental
007	Karachi, Sindh	Chest pain, Breathing Difficulty, and unconsciousness Known case of Hypertension and Cardiac Issues	65	MI	Sinovac & 1st Dose	1166	Died	Coincidental
008	Attock, Punjab	Fever and dizziness, Breathing difficulty, Severe headache, and Vomiting. Known case of Diabetic mellitus and Hypertension	54	MI	Astra Zeneca & 1st Dose	1166	Died	Coincidental
009	Gujranwala, Punjab	Severe vomiting and Chest discomfort & Pain unconsciousness Known case of Hypertension, DM, and ischaemic Heart disease	56	MI	Sinovac & 1st Dose	Deputy National Program Manager, Federal EPI-Islamabad	Died	Coincidental
010	Multan, Punjab	High-Grade fever, Severe diarrhoea, Left Shoulder Pain, Breathing Difficulty	53	Acute ischaemic Event	Sinovac-1st Dose	Journalist through AEFI Consultant	Died	Coincidental
011	Lahore, Punjab	Fever and Severe headache and Vomiting. Known case of Diabetic mellitus and Hypertension	42	CVA	AstraZeneca-1st Dose on 12 May-2021	Deputy National Program Manager, Federal EPI-Islamabad	Died 25 May-2021	Coincidental
012	Sheikhupura, Punjab	Chest Pain, Vomiting and vertigo, loss of memory Know case renal failure and on weekly dialysis	63	MI	AstraZeneca-1st Dose	1166	Alive	Coincidental
013	Khushab, Punjab	Suddenly collapsed on 11- June	58	MI	Sinovac 2nd dose on 31-May-2021	Social Media	Died 11 June-2021	Coincidental
014	DI Khan, KPK	Today information was shared on Call: 13th of Ramzan vaccinated, Severe diarrhoea with fever on 1st day of Eid. Visited DI Khan DHQ Emergency and private practitioner. 23 May-2021 breathing difficulty with Fever and Covid- 19 Positive, on oxygen 27th May admitted in Covid-19 Ward Mufti Mehmood Teaching DI Khan. Unfortunately died on 1st June-at 5:30 PM. Known case of cardiac disease (coronary bypass surgery in 2014).	67	Bronchial pneumonia	CanSinoBIO	Reported via email to DRAP; further information is awaited.	Died	Coincidental
015	Karachi, Sindh	Swelling on Face and allergy	Above 50	Skin Allergy	1st-Sinovac	Reported By AEFI Consultant	Alive	Coincidental

(*continued on next page*)

Table 3 (*continued*)

EPID#	Jurisdiction	Adverse Event	Age	Diagnosis	Vaccine type & dose	Reporting Channel	Status	Conclusion by the national AEFI review committee
016	Islamabad, Capital territory	Fever, Body ache, injection site pain, Severe Neck pain, and swelling	50	Subacute Thyroiditis	CanSino 8–06–2021	Self-report to the Deputy National Program Manager, Federal EPI-Islamabad	Alive	Coincidental
017	Rahim Yar Khan, Punjab	Dizziness & Cold Peripheries, Difficulty in breathing, and nausea Known Diabetic and hypertensive Patient	56	Hypertension and Anxiety reaction	Sinovac-2nd Dose 17–07–2021	Reported by the District AEFI Focal Person	Alive	Coincidental
018	Swabi, KPK	Swelling on Face and eyes. High-grade fever.	39	Skin Allergy	1st Dose Sinovac 27–07–2021	National Program Manager, EPI-Islamabad	Alive	Coincidental
019	Lahore, Punjab	Vomiting, Discomfort, Severe Chest pain tachycardia leads to MI	20	MI and Cardiac Arrest	1st Dose Sinopharm 29–07–2021	Covid-19 consultants and report on Social Media	Died	Causality Assessment under Process
020	Islamabad, Capital territory	Unable to close eyes, Loss of feeling in the face. Headache, Tearing in right eye, Drooling, and Loss of the sense of taste on the front two-thirds of the tongue.	39	Bell's Palsy	2nd Dose Sinovac 23-08–2021	Consultant physician and directly reported by the patient to the National Consultant AEFI	Alive	Under investigation
021	Karachi, Sindh	Severe Headache and Vomiting and breathing difficulty and collapsed	37	Under investigation	1st of Sinovac 30-08–2021	Reported Through Social Media	Died	Under investigation
022	RYK, Punjab	Vomiting, Discomfort, Severe Chest pain and vomiting. Known case of DM, HTN, Chronic Smoker	67	IWMI	2nd Dose AstraZeneca 4–08–2021	1166 helpline	Alive	Coincidental
023	Islamabad	Vomiting, Vertigo, and headache and suddenly semi-unconsciousness	15	Fainting	1st Dose Pfizer- 2nd October-2021	AEFI Focal person ICT	Alive	Under investigation
024	Vehari, Punjab	Mild Fever on first day then suddenly collapsed and died	16	Cardiac arrest	1st Dose Pfizer 29 September-2021	Social Media and District Team	Died	Under investigation

beyond three days post-immunisation. Most adverse events (~70%) were reported after the first dose, whereas events reported after the second dose constituted 30% of total adverse events.

The overall rate of AEFIs reports in Pakistan ranged from 0.27 to 0.79 per 1000 for various Covid vaccines, and this was significantly lower than the rates observed in UK (~4 per 1000). As per MHRA Covid vaccines pharmacovigilance system, there were ~3 AEFI per 1000 reported for Pfizer, ~5 AEFI per 1000 for AstraZeneca and 6 AEFI per 1000 for Moderna (Fig. 2A) [6]. The underreporting of vaccine adverse events in Pakistan was also confirmed by reviewing post-immunization thrombocytopenia case reports. In United Kingdom, 438 cases of major thromboembolic events with concurrent thrombocytopenia (thrombotic thrombocytopenia syndrome, TTS) were reported by MHRA following Covid vaccine AstraZeneca. There were 159 confirmed cases of cerebral venous sinus thrombosis (CVST) and 279 additional major thromboembolic events with concurrent thrombocytopenia, with overall incidence rates of 15.6 cases per million doses administered [6]. In contrast, there were no cases of TTS or CVST reported in Pakistan following AstraZeneca or other viral vector vaccines. Vaccine induced thrombocytopenia (without major thrombosis) was also acknowledged by the European Medicines Agency as a common side effect of the Covid vaccine [7], i.e., occurrence between 1 in 100 to 1 in 10 (1 to 10%) following immunization. Interestingly, there was no case of thrombocytopenia following Covid immunization reported from Pakistan at the time of this analysis.

The lower rates of AEFIs in Pakistan may be driven by various factors, mainly due to an overall underreporting of cases as the national pharmacovigilance system is still under infancy in Pakistan as compared to the well-established MHRA's Yellow Card reporting system that has been in place in UK for a long time. The other factors include hesitancy amongst healthcare professionals for reporting AEFIs amidst fears of antivax behaviours and poor vaccine uptake amongst Pakistani diaspora, lack of training and awareness amongst healthcare professionals at large about significance of pharmacovigilance, and overall, a lack of culture of AEFIs reporting within healthcare settings in Pakistan. Most healthcare professionals in Pakistan do not feel the necessity to report adverse events that are already listed on a product label, or the minor adverse effects that are usually anticipated post-vaccination, such as injection site pain, soreness, fever, or chills etc. Since these events corroborates most of the AEFIs following Covid vaccines, therefore, lack of motivation or the need to report these events may be a significant contributory factor responsible for the overall low rates of AEFIs in Pakistan. The Drug Regulatory Authority of Pakistan has conducted series of training sessions and seminars for healthcare professionals across the country to raise awareness on AEFIs reporting. In particular, several training and awareness sessions were conducted during pandemic to promote Covid-19 vaccine national pharmacovigilance system amongst healthcare professionals and public in general to facilitate AEFI reporting. The new 'Med Safety' mobile phone application was also launched during pandemic in Pakistan and advertised through conventional and social media platforms to raise public and professional awareness at large to encourage AEFIs reporting for all new drugs including Covid-19 vaccines.

Conclusions

As of 30th September 2021, there have been 39,291 ADRs reported across Pakistan, where most of the AEFIs were reported after the first dose (n = 27,108) and within 24–72 h of immunization (n = 27,591). There were 24 serious AEFIs that have been reported to date from over 94 million doses administered. Most common AEFIs included mild to moderate effects such as fever or shivering (35%) followed by injection-site pain or redness (28%), headache (26%), nausea/vomiting (4%), and diarrhoea (3%). The overall rate of AEFI reporting in Pakistan ranged between 0.27–0.79 per 1000 doses administered that was significantly lower than the rates in countries with well-established pharmacovigilance system, for instance in UK ~4 cases per 1000 were reported. Overall, Covid vaccines were well tolerated and no significant cause for concern was flagged up during Pakistan's Covid vaccine surveillance system concluding Covid vaccine benefits outweighed risks.

Acknowledgements

Vaccine pharmacovigilance data was received from Pakistan National Pharmacovigilance Centre (PNPC), Drug Regulatory Authority of Pakistan (DRAP), who have been acknowledged with thanks for their cooperation and providing access to the vaccine surveillance data. Authors would like to thank all staff at the PNPC and DRAP who were involved in active monitoring and surveillance of Covid vaccine safety in Pakistan and enabled the data for this manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. No funding was received to prepare this manuscript. AR was the Director for the Pakistan National Pharmacovigilance Centre, Drug Regulatory Authority of Pakistan (DRAP) at the time of writing this analysis.

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